

Pretesting

Wenchao Ma
University of Minnesota



Pretesting

- Pretesting is the process of evaluating the survey procedures in advance to assess whether they are going to cause any problems for respondents and interviewers and whether the survey is meeting its intended objectives (Presser et al., 2004)



Why Pretesting?

- Gather others' input to gain new perspective
- Develop greater understanding of and empathy with desired respondents
- Ensure questions are clear, comprehensible, and relevant to respondents (Collins, 2003)
- Confirm that response options are sufficient and appropriate and that respondents can use them appropriately and consistently (Collins, 2003)



Why Pretesting?

- Explore whether questions have particular forms of validity, such as face validity
- Make initial assessment regarding the reliability of questions
- Anticipate how the resulting data might best be analyzed
- Test the mechanics of the survey in a real-world setting



Methods for Item Review in Pretest

Qualitative methods



- ☐ Cognitive Interview
- ☐ Expert review
- ☐ Respondent Debriefing
- ☐ Focus Group Interview

Quantitative methods

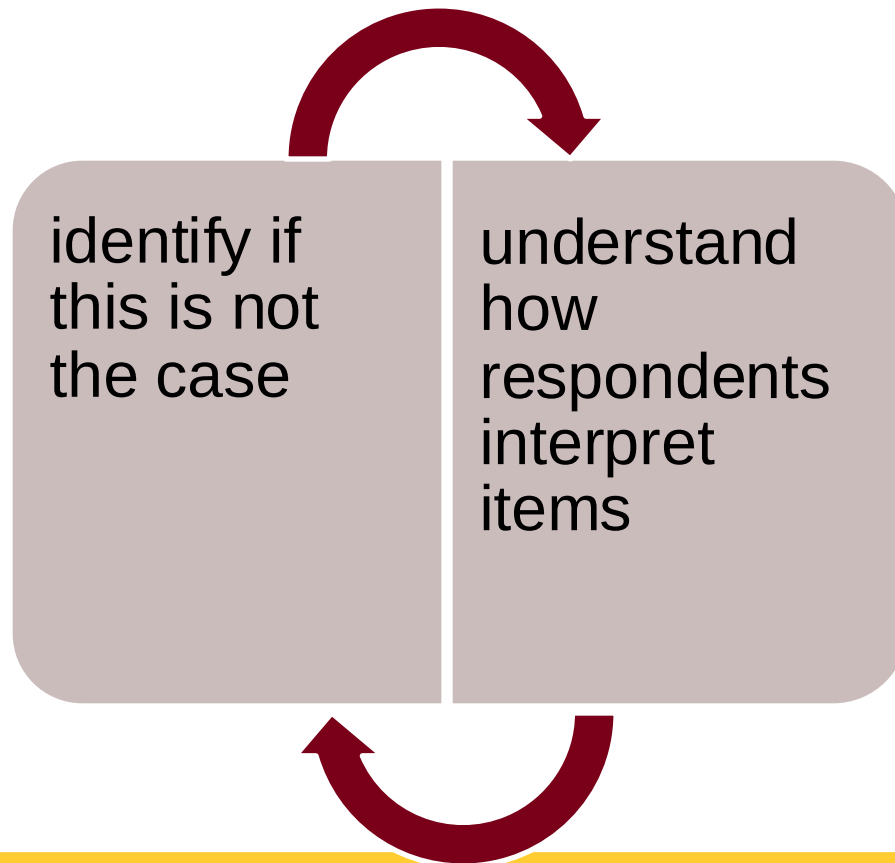


- ☐ Behavior Coding
- ☐ Paradata



Goal of item review

We assume respondents interpret items in the manner intended by the survey developer



An example

Considering your work and commute schedules, would you say the division of your housework is fair?

- 1) No, I do way too much of the work.
- 2) No, I do somewhat more than I should.
- 3) No, my partner does way too much of the work.
- 4) No, my partner does somewhat more than she/he should.
- 5) Yes, it is fair enough.



Pretest sample size

- Pretest the survey on at least 12 to 50 respondents from the target population



Cognitive Interview

- The objective of cognitive interviewing is
 - to diagnose item ambiguities in language and response scale understanding, and
 - to identify challenges respondents experience and mental operations that may be inappropriate in responding to items.
- CI Method Protocols
 - Think-aloud interviews
 - Verbal probing



Cognitive Interview

- Conduct one-on-one in a laboratory setting or on-line
- Ask respondents to verbalize whatever comes to their mind as they respond to each of the items presented to them
- Interview needs to be audio or video recorded
- Data generated by think-aloud are often not formally analyzed, but results can be quantified by counting the number of problems encountered with a question.



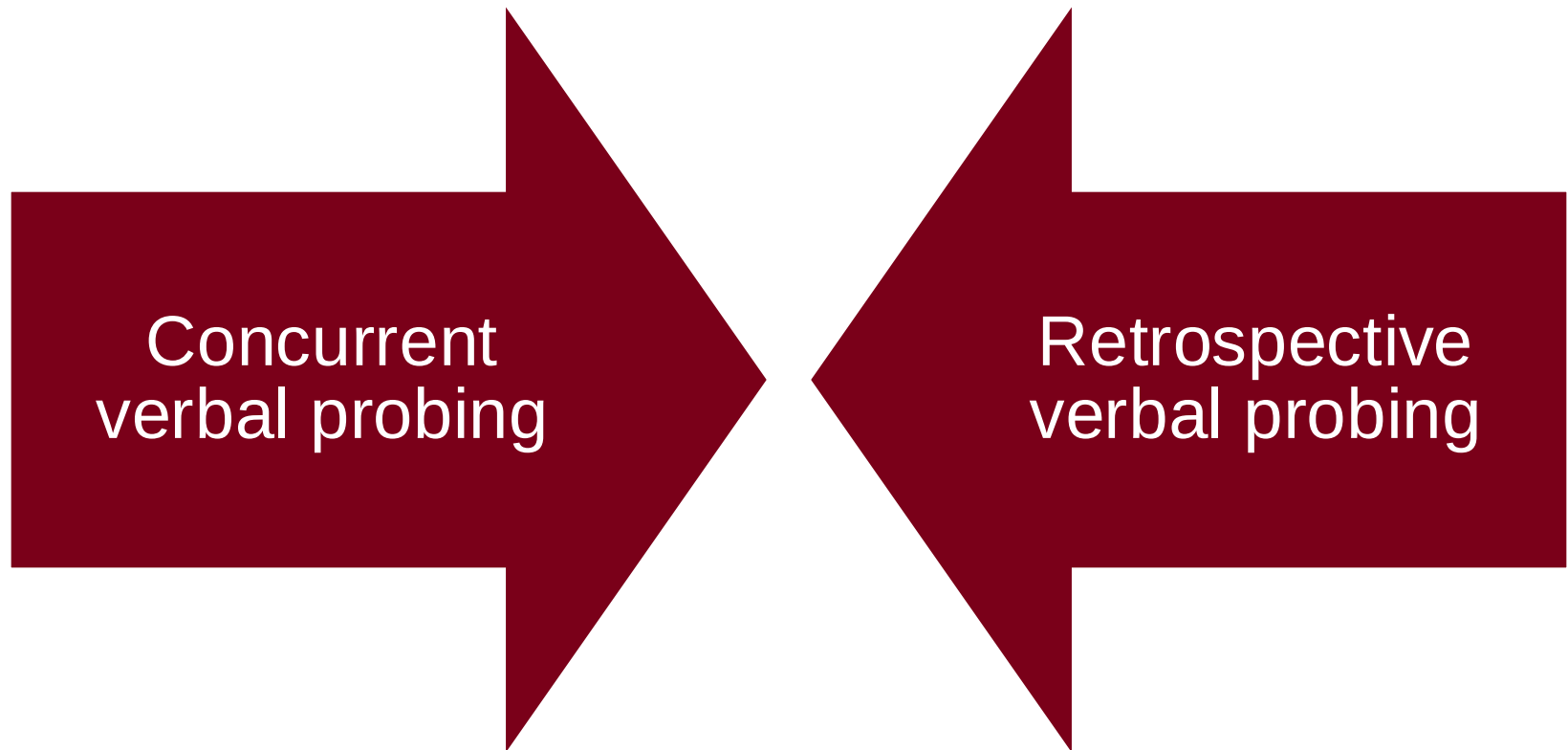
Think-aloud interviews

A sample instruction

Thinking aloud may be new and unfamiliar to you, but please know there are no wrong answers. I am only interested in knowing what is going through your mind. Before we begin the actual interview, I'd like to ask you a 'warm-up' question to introduce you to the think aloud process. Feel free to comment or ask questions as you read and answer the question. I did not write the questions and you will not hurt my feelings, no matter what you say. There are no wrong answers. What questions do you have before we get started?



Verbal Probing Interview



An example of concurrent verbal probing

- **Interviewer** (asks targeted survey question) In the past 7 days, on how many days did you walk for exercise.
- **Subject** Uh, three.
- **Interviewer** (probe) Tell me about those times.
- **Subject** Well, I probably went to the gym twice, and I also walked to the store yesterday. I'm assuming you want to include that – it's uphill and I did get some real exercise carrying those bags.
- **Interviewer** (probe) How sure are you that these were all in the past 7 days?
- **Subject** I don't know – I usually go to the gym twice a week. I don't remember anything different about this last week, you know, in that regard.



An example of retrospective verbal probing

- **Interviewer:** Earlier I asked you the question: In the past 7 days, on how many days did you walk for exercise? Do you remember how you answered that?
- **Subject** Uh, yeah, I think I said three times.
- ...



Verbal Probing Interview – sample probing questions

- Asking respondents how they arrive at their answers.
 - “Why did you choose [‘infrequently’]?”
 - “How did you come up with the number [4]?”
 - “What process did you go through to decide on [‘often’]?”
 - “What went through your head when you choose [‘somewhat agree’]?”
 - “Why did you select the response you did?”



Verbal Probing Interview – sample probing questions

- Asking respondents to paraphrase their understanding of the question or part of the question.
 - “In your own words, what is the question asking?”
 - “What does the term...mean to you in this question?”
 - “What does ‘moderate to intense’ mean to you?”
 - “Can you repeat the question in your own words?” “What terms need to be defined?”
 - “What terms need examples to help explain them?”



Verbal Probing Interview – sample probing questions

- Asking respondents how confident they are that they can give an accurate answer.
 - “How sure are you of your answer?”
 - “Are you sure about your answer?”



Verbal Probing Interview – sample probing questions

- Asking respondents for any uncertainties or confusion they had about what the appropriate answer was.
 - “Is there any specific part of this item (including the statement and options) that you are not comfortable with?”
 - “Is there anything in this item that confuses you?”
 - “Is there anything that didn’t feel right or missing?”
 - “Was there anything that seemed redundant or should be deleted?”



Verbal Probing Interview – sample probing questions

- Asking respondents for suggestions for improving items.
 - “How would you improve the item?”
 - “What can be done to this item to make it better?”
 - “What needs to be expanded?”



Think-Aloud vs Verbal Probing

Think-Aloud

- (+)Fewer interruption
- (+)Minimal interviewer training
- (-)depends on respondents

Verbal Probing

- (+)Less dependent on respondents
- (-)Potential bias
- (-)Respondents may not remember in retrospective verbal probing



Cognitive Interview – Data Analysis

- Data from cognitive interview are text
- We use data to understand
 - which items and which words in items are confusing to interviewees,
 - which items or words in items are misinterpreted by interviewees, and
 - how to revise the items so they are clearer.



Expert Review

- Select three to six experts—qualifying as an expert requires familiarity with the topic of the survey and/or survey methodology.
- Experts conduct their reviews independently but could review items as a group.
- A rating form is often used to record reviews and reviewers are often asked to read items for bias and for technical flaws



Checklist for Quality Question Design (Robinson & Leonard, 2018)

- assist the survey researcher in designing an effective survey instrument that will elicit a rich, robust, and useful dataset
- This tool will be of greatest use if the following has already been determined:
 - That a survey is indeed the appropriate tool to meet data collection needs
 - The research or evaluation questions and information
 - The population or sample of desired respondents
 - How the survey will be administered: online, paper, by phone, in person, or some combination of these
- Going through the checklist for each survey question



Checklist for Quality Question Design (Robinson & Leonard, 2018)

Checklist Item	Yes	No	N/A	Notes
Question type is appropriate:				
Question type will result in the type of data needed given research question(s)	☐	☐	☐	
Researcher is prepared to analyze results from this type of question	☐	☐	☐	
Question design is centered around respondents:				
Question asks for information respondent is likely to know, or respondent will have access to the information needed	☐	☐	☐	
Question focuses on respondent and respondent experiences (and not what respondent thinks about others)	☐	☐	☐	
Researcher can anticipate how a respondent might answer the question	☐	☐	☐	



Question Appraisal System

- Question Appraisal System QAS-99



Expert Review

- Use those checklist as a starting point
- All surveys are different – apply the checklist in light of the survey context



Respondent Debriefing

- Respondent debriefing involves informal or specific, structured, and standardized follow-up questions
- Goal is understanding how respondents interpreted the survey items after the respondent completes the survey
- Can be conducted individually, or use a standardized form administered on-line



Focus Group Interview

- Sometimes used to generate content/ideas for item writing
- Can also be used to evaluate a set of potential items



Focus Group Interview

- Prior to the focus group, participants are usually recruited
- The ideal size for a focus group is generally between six and twelve people.



Focus Group Interview - Procedure

- At the focus group itself, the moderator begins with an introduction that should include the following:
- explaining the purposes of the focus group
- laying down some basic ground rules to encourage everyone to participate in the discussion
- reassuring the participants about the voluntary and confidential nature of their participation
- introducing the moderator and any co-moderators and explaining how and why these group members were invited to participate
- stating the purpose of note taking and recording.



Focus Group Interview – sample questions

- **Understanding the Questions**
 - How do you interpret this question: “[Insert survey question]”?
 - Is there any part of this question that is unclear or confusing?
 - What do you think the question is trying to measure?



Focus Group Interview – sample questions

- **Relevance and Context**

- How relevant do you think this question is to your experience?
- Can you think of any situations where this question would not apply?
- What additional context or background information would help you understand this question better?



Focus Group Interview – sample questions

- **Completeness of the Questions**

- Do you feel that this question covers all the necessary aspects of the topic?
- Are there any important aspects that you think are missing from this question?
- If you could add a follow-up question to this, what would it be?



Focus Group Interview – sample questions

- **Response Options**

- Are the answer choices provided adequate for this question? If not, what would you suggest?
- How would you feel about the format of the response options? (e.g., Likert scale, multiple choice, open-ended)
- Is there any response option that you find confusing or misleading?



Focus Group Interview – sample questions

- **Tone and Sensitivity**

- How does this question make you feel? Does it seem respectful and appropriate?
- Are there any words or phrases that you think may be off-putting or insensitive?
- How would you suggest adjusting the tone to be more inclusive or relatable?



Behavior Coding

- BC was initially implemented to monitor interviewer performance, by
 - identify instances where the interviewer has difficulty asking questions,
 - the respondent has difficulty answering them, or both
- BC in its normal form is only applicable to interviewer-administered instruments
- BC is usually applied at a later point in the development of a survey questionnaire than is cognitive interviewing
- BC is often conducted a spart of a conventional pretest (i.e., a dry run)



Behavior Coding - procedures

- Development of coding scheme
- Development of system for recording and processing results.
- Collection of audio data.
- Compilation of codes, for each item.
 - A summary table can be produced which displays frequencies of each coded item, over all interviews
 - A criterion can be set for problem frequencies to identify particularly problematic items—typically around 15% as suggested by Fowler (2011)



Behavior Coding

- Common “interviewer codes” include
 - interviewer reads question exactly as worded,
 - makes minor changes, or
 - makes major changes.
- Common “respondent codes” are
 - respondent asks for clarification,
 - respondent gives an inadequate answer,
 - respondent asks interviewer to repeat the question,
 - respondent interrupts,
 - respondent provides a qualified answer (indicating uncertainty), or
 - respondent provides an adequate answer.



Paradata

- A variety of possible measures of auxiliary data regarding survey response behavior such as
 - response time
 - eye movement
 - logfile data
- Response latencies that are systematically longer could be used as a flag for item review

